

ACM RECSYS 2026 · RECSYS CHALLENGE WORKSHOP

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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47,071

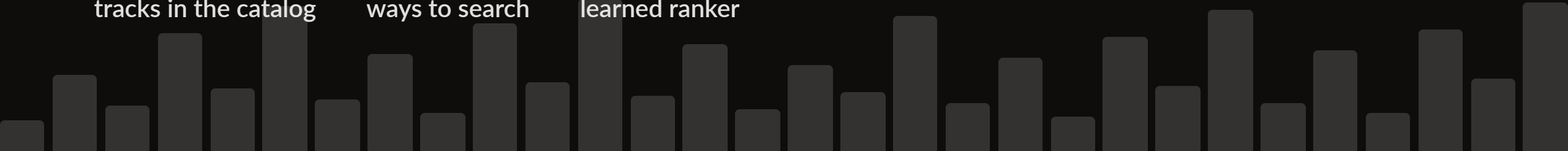
tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of approximately 20 vertical bars of varying heights in a dark gray color.

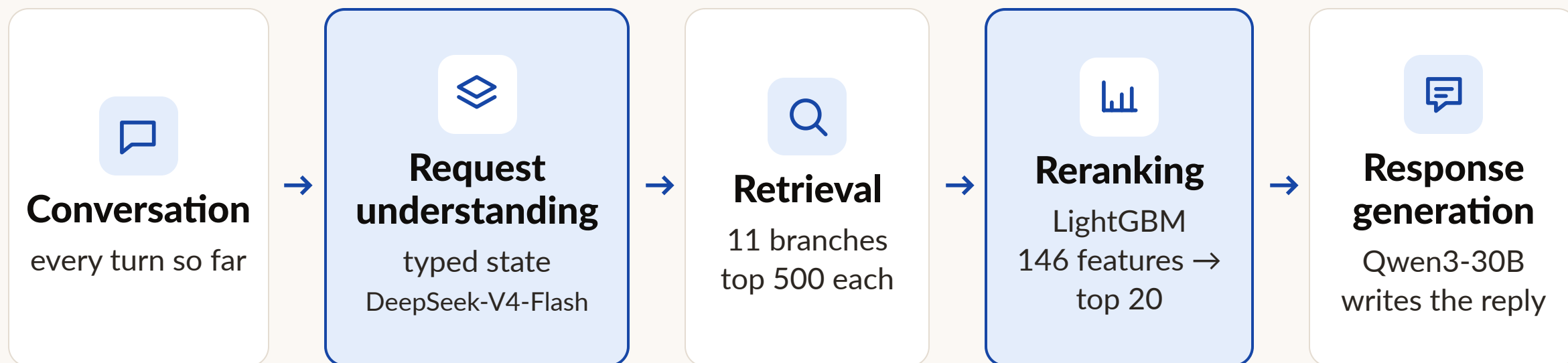


Contents

1	How it works	the state, retrieval, reranking, the reply
2	How it scored	the official results
3	Why we did not score higher	training, missing data, labels, ranking, the reply
4	Conclusion	what we showed, and questions

System overview

APPROACH



Request understanding → conversation state

APPROACH

WE HAD JUST PLAYED



Ryan Adams & The Cardinals — “Let Us Down Easy”

DeepSeek
V4-Flash



LISTENER

...more tracks with that kind of emotional depth and acoustic focus, maybe branching out to other singer-songwriter or alt-country artists?

CONVERSATION STATE

REQUEST TYPE

new artist

AVOID

Ryan Adams & The Cardinals

MOOD

introspective,
emotional depth

SOUND

acoustic-driven

GENRE

singer-songwriter, alt-country

DeepSeek-V4-Flash reads **every turn so far** and writes a typed **state**: “other artists” after a Ryan Adams track becomes an artist to avoid, and the mood carries over.



What the state can hold

APPROACH

THE REQUEST

- request type
- intent mode
- routing flags

WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

WHAT IT IS ABOUT

- lyrical theme

BOUNDS

- release-year window
- process constraints



How the state drives retrieval

APPROACH



The request

request type · intent mode · routing flags



Which branches fire, and how much each one weighs in the merge

routing weights

pivot skips anchors

gates discography



What they like

liked tracks · named artists and tags



Search near liked tracks; match named entities

anchor centroids ×3

BM25 fields

artist discography



What it sounds like · is about

mood · genre · sonic · lyrical theme



The text, audio and lyrics queries

dense text ×3

CLAP text→audio

lyrics view



What they ruled out

rejected tracks and artists



Dropped from every branch before the ranker

hard drop

same-artist demote



Bounds

release-year window · date filters



Mask out-of-range tracks; boost the era

candidate mask

era-popularity lookup

Retrieval — eleven branches

APPROACH



×1

BM25 + tag resolver

keywords and resolved tags
from the state



×3

Dense text

Qwen3 embeddings of lyrics,
metadata, attributes



×1

CLAP text → audio

the sonic description,
matched to audio



×3

Anchor centroids

near liked tracks: cover art,
audio, listening



×1

User-CF centroid

this listener's own BPR
vector



×2

Catalog lookups

the artist's discography; era
popularity

Each branch returns its **top 500**; the ranker sees their union.



Reranking

APPROACH

Pooled candidates
top 500 per branch



LightGBM ranker
146 features per candidate



Top 20
the scored list

WHAT IT LEANS ON · SHARE OF SPLIT GAIN

Bi-encoder cosine

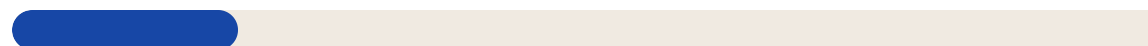
· conversation vs track card



59.0%

Per-branch rank and score

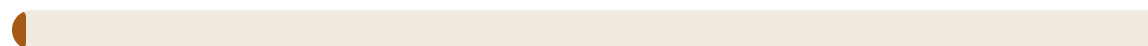
· 66 features



11.5%

Constraints and rejections

· 13 features



0.7%

One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



Response generation

APPROACH

INPUTS

Conversation

"...it feels a bit too serious. I'm looking for something more uplifting and positive..."

Latest state

Mood **uplifting, positive**

Avoid **serious**

Top-ranked track

Track **Röyksopp – "Happy Up Here"**

Tags **cheerful, hopeful, electronic...**



Qwen3-30B

temperature 0 • one
pass

1-2 sentences about
the top track only



OUR REPLY

"Perfect – 'Happy Up Here' by Röyksopp is exactly the kind of uplifting, cheerful electronic track you're looking for. With its bright synths and hopeful groove, it's designed to lift your spirits and make you smile."

How it scored

RESULTS

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



Why we did not score higher

WHAT IS NOT IDEAL

- 1 Why we trained on the dev set
- 2 No data from outside the catalog
- 3 Inconsistent labels
- 4 Extracted, but not acted on
- 5 Gaps in our response generation



Why we trained on the dev set

LESSONS



State extraction: 1 paid LLM call per turn

to read each turn into a state



Too costly to extract state for 121.6k training turns

so no ranker features there



Trained on the 8,000 dev turns

the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.

No data from outside the catalog

WHAT IS NOT IDEAL

A tempo and a key

“exactly 126.70 bpm, in G minor”



the catalog has no tempo or key field
no branch can match on either

A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”
no film or TV links in the catalog

The state was right both times. The gap is **knowledge, not understanding**.

Inconsistent labels — “from a different artist”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Bonobo — “Cirrus”

LISTENER, NEXT TURN

...something with a similar chill, electronic vibe, maybe with a subtle beat, but **from a different artist?**

ANSWER KEY FOR THAT TURN



Bonobo — “Jets”

MOVES TOWARD GOAL

NEXT TURN

“Jets” is cool, but I was actually hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

17% of training turns keep the artist the listener just asked to leave
Two LLM judges re-scored all 113k labelable turns; Opus broke ties.

⊗ Extracted, but not acted on

WENT WRONG

*"I gotta break away from **Deltron** and **Del** completely"*

✓ STATE EXTRACTED
avoid Deltron 3030, Del

⊗ OUR TOP 20
10 of 20 by them — from rank 6
retrieval kept them: names resolved to the wrong catalog IDs

*"branching out to **other** singer-songwriter artists"*

✓ STATE EXTRACTED
avoid Ryan Adams & The Cardinals

⊗ OUR TOP 20
10 of 20 by Ryan Adams — rank 1
retrieval kept them: recorded as "avoid exact", not a rejection

*"still focusing on **Juanes**... haven't found the depth I'm after"*

✓ STATE EXTRACTED
rejected Juanes

⊗ OUR TOP 20
7 of 20 by Juanes — rank 1
retrieval kept them: a soft rejection only feeds ranker features

RANKER SPLIT GAIN Bi-encoder cosine

59.0%

Constraints and rejections

0.7%



Gaps in our response generation

WHAT IS NOT IDEAL

The track is not in the catalog, so retrieval could not return it — the reply presents our substitute as if it were the request

1

"I wanted 'Watercolors' by Pat Metheny specifically."
"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

Our pick broke the listener's constraint — the reply recommends it anyway

2

*"...something from a **different artist** this time"*
"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).



Conclusion

WHAT WE SHOWED



A typed conversation state

- one LLM call per turn
- request, likes, rejections, sound, bounds



Retrieval → ranking → reply

- eleven retrieval branches
- a learned ranker
- a reply about the top track



Gaps in our system

- inconsistent labels — 17% keep a rejected artist
- trained and checked on the same turns
- rejections not acted on
- no knowledge outside the catalog
- the reply was never checked

Thank you — questions?



Paper, poster, slides, code, models, data
npatta01.github.io/music-crs-2026

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Appendix

TYPE A NUMBER + ENTER • OR G TO SEARCH

- 1 Six request types, six shapes of state
- 4 The reranker
- 7 Failure 3 — the better track was already in the pool
- 10 The target often ignores the request

- 2 Routing flags decide which branches fire
- 5 Down-weighting contradicted turns
- 8 How the relabelling worked

- 3 The eleven branches
- 6 The b1 bi-encoder
- 9 What the relabelling found



Six request types, six shapes of state

APPROACH

REQUEST TYPE	A REAL TURN	WHAT THE EXTRACTOR PULLED OUT
<code>exact_track</code>	"I wanted <i>Watercolors</i> by Pat Metheny specifically."	track = <i>Watercolors</i> must_use · artist = Pat Metheny must_use
<code>exact_album</code>	"Which song from Peter Doherty's <i>Grace/Wastelands...</i> "	artist + album must_use · facet "best tracks of 2009"
<code>same_artist</code>	"Got another track known for his guitar work?"	artist must_use · prior track do_not_use · facets: guitar work, melodic complexity
<code>new_artist</code>	"...branching out to other singer-songwriter artists."	artist = Ryan Adams & The Cardinals do_not_use · facets: acoustic, introspective
<code>attribute_search</code>	"...a really striking, abstract cover art, bold colors and shapes."	three visual facets · routing flag image_or_visual_search
<code>hidden_target</code>	"A Lil Wayne song from an experimental phase."	artist must_use · facet "experimental" · flag hidden_target_search

The extractor read all six of these correctly. Across the whole Blind-B run only **3 of 80 turns** were diagnosed as a state-extraction miss — reading the conversation was almost never the problem.



Routing flags decide which branches fire

APPROACH

FLAG	OPENS
exact_entity_probe	catalog lookups – discography, album
lyric_search	the Qwen3 lyrics view
feature_articulation	CLAP text→audio on the sonic description
image_or_visual_search	the gated SigLIP-2 cover-art branch
hidden_target_search	widens the pool; suppresses exact pins

A REAL VISUAL REQUEST

“I’m looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets,
`image_or_visual_search = true`

AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.

The eleven retrieval branches

APPROACH

BRANCH	SPACE	QUERY
BM25 (fielded)	lexical	state text + resolved tags
Dense text × 3	Qwen3 views	state-built strings
CLAP text→audio	audio	sonic description
Anchor centroids × 3	SigLIP-2/CLAP/CF	liked-track centroids †
User-CF centroid	CF-BPR	the listener's own vector
Lookups × 2	catalog	discography; era popularity

GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.



What the ranker leans on

APPROACH



Share of the ranker's split gain, by feature family. One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



Down-weighting contradicted turns

APPROACH

Every turn starts at		1.0
Next turn rejects the artist		× 0.6
Next turn rejects the track		× 0.3
Goal note: did not help		× 0.3
Never below		0.2

WHY

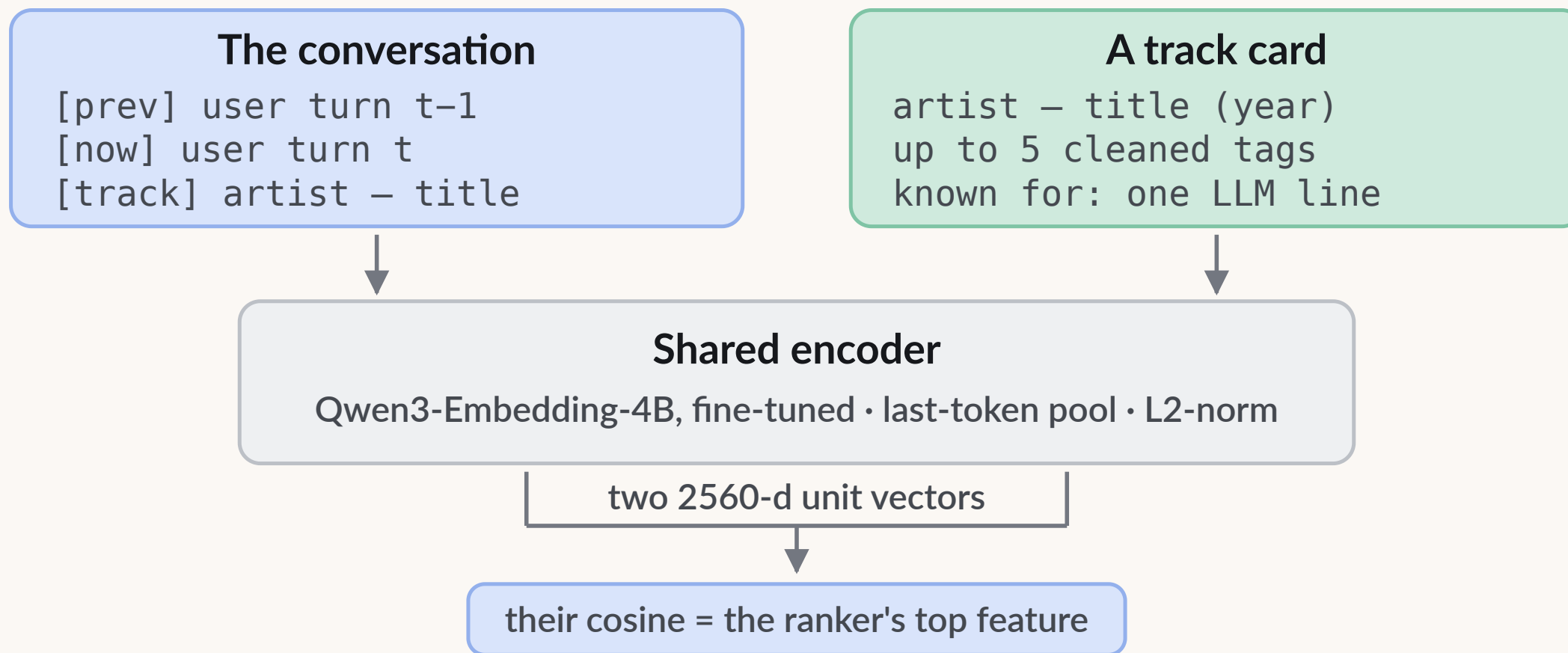
The label is whatever played next. When the **next** turn says it was wrong, that label counts for less in training.

Chosen by hand, never tuned. Goal notes shape these weights but are never a ranker input.



The bi-encoder behind that top feature

APPROACH





Failure 3 — the better track was already in our pool

ANALYSIS

WHAT THEY ASKED FOR	WE SERVED AT RANK 1	ALREADY IN OUR POOL	POOL RANK
nu-jazz, downtempo, female vocals — <i>not</i> Parov Stelar	Parov Stelar — Booty Swing	St Germain — What You Think About...	34
new underground hip-hop, not Deltron	Dead Prez — Hip Hop	Souls Of Mischief — Make Your Mind Up	31
hardcore, not classic hip-hop	Dead Prez — Hip Hop	Papa Roach — Getting Away With Murder	49
that carol, but not Beegie Adair again	Beegie Adair — The Christmas Song	Frank Sinatra — Have Yourself A Merry Little Christmas	63

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.

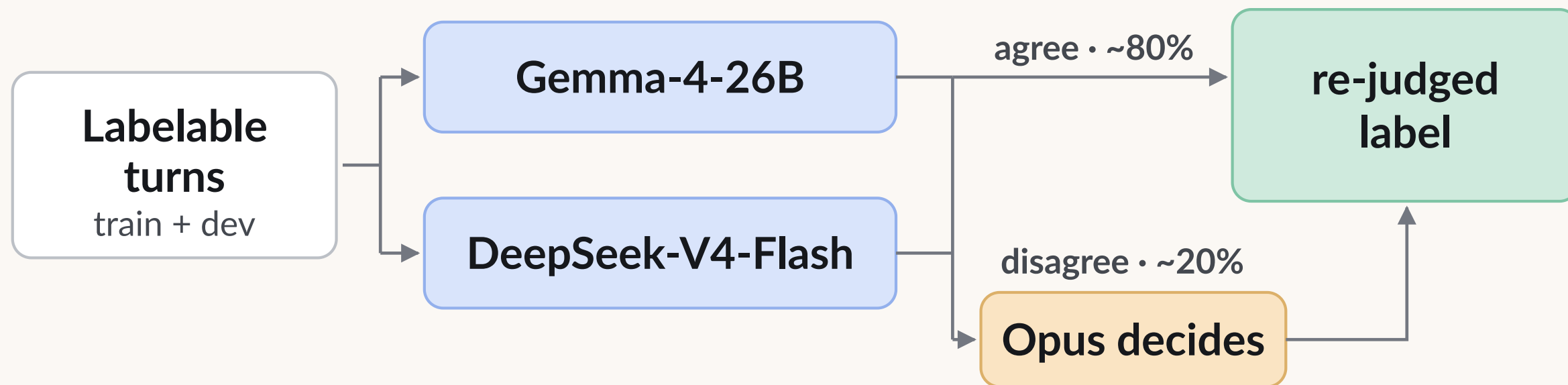


How the LLM relabelling worked

ANALYSIS

The answer keys come from the simulated recommender's picks. To see how often they actually fit the request, we had **two LLM judges re-score every labelable turn**.

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



Agreement split measured on the training turns.



What the relabelling found

ANALYSIS

17%

of training turns keep the artist the listener had just asked to leave

18,222 turns — both judges or the arbiter marked the label as artist anchoring

113,393

turns re-judged

every labelable train and dev turn

Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix — so we kept the originals.



The target often ignores the request

ANALYSIS

LISTENER · JUST PLAYED BONOBO

...something with a similar chill, electronic vibe, but **from a different artist?**

ANSWER KEY



Bonobo — “Jets”

MOVES TOWARD GOAL

NEXT TURN

“Jets” is cool, but I was hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

The answer key repeats the artist the listener just asked to leave. Our ranker **anchors** the same way — we think it learns it here.